USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [22CSE104]

# SECOND SEMESTER B.Tech EXAMINATION, JUNE - 2023

## COMPUTER ORGANIZATION AND ARCHITECTURE

**Time: 3 Hours** **Maximum Marks: 100**

**Instructions: i.** Missing data may be suitably assumed.

**ANSWER ALL QUESTIONS**

**PART - A**

**5 X 2=10**

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|---|---|---|---|---|---|
| **1.** | **a.** | Which type of gate can be used to implement any logic function?<br>**i.** AND gate **ii.** OR gate **iiii.** NOT gate **iv.** NAND gate | L-3 | CO1 | **02** |
| | **b.** | Which type of shift micro operation shifts the bits to the right and fills the vacant bits with zeros?<br>**i.** Logical right shift **ii.** Logical left shift<br>**iii.** Arithmetic right shift **iv.** Arithmetic left shift | L-2 | CO6 | **02** |
| | **c.** | Which type of computer architecture incorporating a large number of powerful instructions?<br>**i.** RISC **ii.** CISC **iii.** MIPS **iv.** ARM | L-2 | CO6 | **02** |
| | **d.** | Which technique allows a device to transfer data directly to and from memory without CPU intervention?<br>**i.** Priority Interrupt **ii.** DMA<br>**iii.** Pipelining **iv.** Vector Processing | L-2 | CO5 | **02** |
| | **e.** | What is the purpose of cache memory in a computer system?<br>**i.** To store frequently accessed instructions and data for faster access<br>**ii.** To provide long-term storage for programs and data<br>**iii.** To facilitate communication between the CPU and peripheral devices<br>**iv.** To perform arithmetic and logic operations | L-2 | CO2 | **02** |
| | | **PART - B** | | | **6 X 5=30** |
| **2.** | **a.** | Define the following terms: CPU, memory, control unit, and bus structures. | L-2 | CO1 | **05** |
| | | **OR** | | | |
| | **b.** | Explain the concept of functional units in a CPU. Provide examples of different functional units. | L-2 | CO1 | **05** |
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| **3.** | **a.** | Draw hardware implementation of arithmetic addition and subtraction operation. Explain the steps involved in the circuit design. | L-3 | CO3 | **05** |
| | | **OR** | | | |
| | **b.** | Evaluate the importance of error detection codes in data representation and transmission. | L-3 | CO4 | **05** |
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| **4.** | **a.** | Discuss the working principles of floating-point arithmetic operations. | L-2 | CO3 | **05** |
| | | **OR** | | | |

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| | b. | Discuss any five addressing modes in detail with neat diagrams. | L-2 | CO2 | **05** |
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| **5.** | **a.** | Define and list the modes of data transfer used in input/output organization. Compare and contrast synchronous and asynchronous data transfer methods. | L-3 | CO5 | **05** |
| | | **OR** | | | |
| | **b.** | Discuss the different levels of memory in a typical memory hierarchy and their characteristics. | L-2 | CO5 | **05** |
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| **6.** | **a.** | Discuss the impact of cache size and block size on cache performance, hit rate, and memory access time. | L-3 | CO2 | **05** |
| | | **OR** | | | |
| | **b.** | Compare and contrast serial communication and parallel processing, highlighting the advantages, disadvantages. | L-3 | CO5 | **05** |
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| **7.** | **a.** | Compare and contrast RISC (Reduced Instruction Set Computer) and CISC (Complex Instruction Set Computer) architectures. | L-3 | CO2 | **05** |
| | | **OR** | | | |
| | **b.** | Evaluate direct mapping, associative mapping, and set-associative mapping and also, discuss their advantages, disadvantages. | L-3 | CO5 | **05** |
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| | | **PART - C** | | **6 X 10=60** | |
| **8.** | **a.** | Compare and contrast multiprocessors and multi-computers by highlighting their differences in terms of performance, scalability, and resource sharing. | L-2 | CO2 | **10** |
| | | **OR** | | | |
| | **b.** | Define logic gate. State the names and symbols of the three basic logic gates. Provide an example of a simple logic circuit that uses each of the three basic logic gates. | L-3 | CO2 | **10** |
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| **9.** | **a.** | Define Data Type. Discuss different types of Number System with a conversion Example. | L-3 | CO1 | **10** |
| | | **OR** | | | |
| | **b.** | Describe the process of different Micro Operations in detail. | L-2 | CO3 | **10** |
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| **10.** | **a.** | Develop a RTL sequence to perform an addition arithmetic operation on two given registers. Explain the RTL sequence and provide the step-by-step execution. | L-3 | CO5 | **10** |
| | | **OR** | | | |
| | **b.** | List and explain the components of the instruction cycle in a CPU. Describe the role of each component in executing a computer instruction. | L-2 | CO3 | **10** |
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| **11.** | **a.** | Define and provide examples of data manipulation instructions in computer architecture. | L-2 | CO1 | **10** |
| | | **OR** | | | |

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| | **b.** | Draw and discuss Daisy-Chaining Priority interrupt handling method in detail. | L-2 | CO4 | **10** |
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| **12.** | **a.** | List various techniques used for Modes of Transfer and discuss their advantages and disadvantages in terms of system efficiency and CPU utilization. | L-2 | CO2 | **10** |
| | | **OR** | | | |
| | **b.** | Draw and describe concept of hierarchical memory organization in detail. | L-2 | CO5 | **10** |
| | | | | | |
| **13.** | **a.** | **i.** Distinguish between a synchronous and an asynchronous data transfer<br>**ii.** Describe DMA operation. State its advantages. | L-2 | CO2 | **05**<br>**05** |
| | | **OR** | | | |
| | **b.** | Evaluate different replacement algorithms, such as LRU and FIFO used in cache memory. | L-3 | CO4 | **10** |